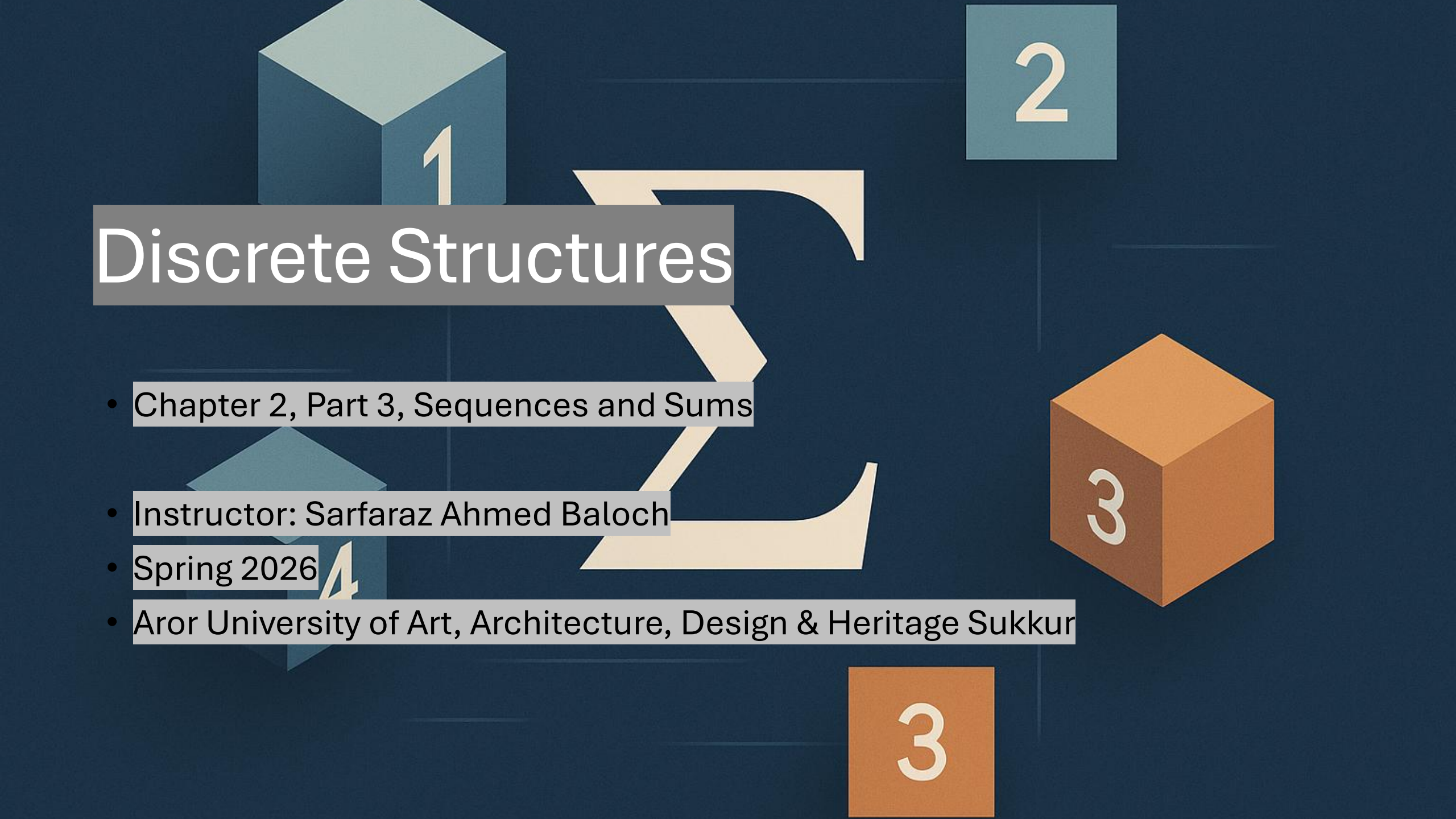


The background is a dark blue gradient. It features several geometric elements: a light blue cube with the number '1' on its side in the top left; a light blue square with the number '2' in the top right; an orange cube with the number '3' on its side in the middle right; and an orange square with the number '3' in the bottom right. A large, stylized, light orange number '5' is positioned in the center. Faint white lines form a grid pattern across the background.

Discrete Structures

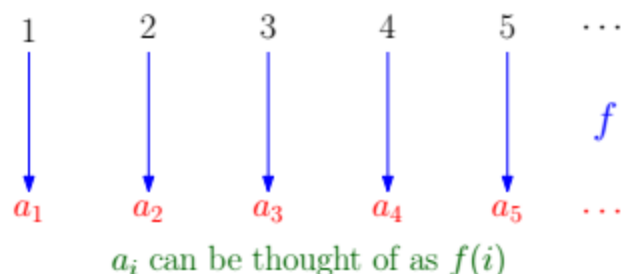
- Chapter 2, Part 3, Sequences and Sums
- Instructor: Sarfaraz Ahmed Baloch
- Spring 2026
- Aror University of Art, Architecture, Design & Heritage Sukkur

Sequences

A sequence is an **ordered list** — could be finite or infinite

- $\{a_i\}$ denotes the sequence
- a_i is the i^{th} term of the sequence

▷ general term



Given a (multi)set S ($n = |S|$), a sequence is a function $f : \{1, 2, \dots, n\} \mapsto S$

- An infinite sequence is a function $\mathbb{N} \mapsto S$
- f represents the order of elements in S ▷ assigns indices to S
- Be careful whether you start indexing from $i = 0$ or $i = 1$

Sequences with Patterns

What is the general term of the following sequences?

1, 2, 3, 4, 5, 6, ...

$$\triangleright a_i = i$$

2, 4, 6, 8, 10, ...

$$\triangleright a_i = 2i$$

1, 3, 5, 7, 9, ...

$$\triangleright a_i = 2i - 1$$

ICP 7-1 2, 5, 10, 17, 26, ...

$$\triangleright a_i = i^2 + 1$$

ICP 7-2 2, 8, 26, 80, 242, ...

$$\triangleright a_i = 3^i - 1$$

Sequences with Patterns

List the first 5 terms of each of these sequences.

The sequence starting with 10 and each term is obtained by subtracting 3 from the previous term

▷ 10, 7, 4, 1, -2

The sequence whose n th term is the sum of the first n positive integers

▷ 1, 3, 6, 10, 15

The sequence whose n th term is $3^n - 2^n$

▷ 1, 5, 19, 65, 211

ICP 7-4 The sequence whose first two terms are 1 and 5 and each succeeding term is the sum of the two previous terms

▷ 1, 5, 6, 11, 17

Geometric Progressions

A **geometric progression** is a sequence of the form

$$a, ar, ar^2, \dots, ar^i, ar^{i+1} \dots$$

where a and r are real numbers

$$\frac{ar^{i+1}}{ar^i} = r$$

- the ratio of consecutive terms is called the **common ratio**
- a is called the **initial term**
- The next term is obtained by multiplying the previous term with r

Arithmetic Progressions

An **arithmetic progression** is a sequence of the form

$$a, a + d, a + 2d, \dots, a + id, a + (i + 1)d, \dots$$

where a and d are real numbers

$$(a + (i + 1)d) - (a + id) = d$$

- the difference of consecutive terms is called the **common difference**
- a is called the **initial term**
- The next term is obtained by adding the previous term with d

Progression Examples

$$1, -1, 1, -1, 1, -1, 1, -1, \dots$$

GP, $a_1 = 1$, common ratio $= -1$, $a_i = (-1)^{i-1}$

$$3, 9, 27, 81, 243, \dots$$

GP, $a_1 = 3$, common ratio $= 3$, $a_i = 3^i$

$$1, 1/2, 1/4, 1/8, 1/16, \dots$$

GP, $a_1 = 1$, common ratio $= 1/2$, $a_i = (1/2)^{i-1}$

Progression Examples

1, 4, 7, 10, 13, 16, ...

AP, $a_1 = 1$, common difference = 3, $a_i = 1 + 3(i - 1)$

6, 2, -2, -6, ...

AP, $a_1 = 6$, common difference = -4, $a_i = 6 - 4(i - 1)$

Progression Examples

For the following sequences, write the type of progression, initial term, common ratio/difference and general term?

ICP 7-5

$1, -2, 4, -8, 16, -32, \dots$

ICP 7-6

$-2.7, 0.1, 2.9, 5.8, \dots$

Some Common Sequences

a_i	First 10 Terms
i^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...
i^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ...
i^4	1, 16, 81, 256, 1296, 2401, 4096, 6561, 10000, ...
2^i	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ...
3^i	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049
$i!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800

Summation

The symbol $\sum$ converts a sequence of numbers into a **sum**

$$\sum_{i=0}^n a_i = a_0 + a_1 + a_2 + \cdots + a_n$$

pronounced as “sum of a_i , where i goes from 0 to n ”

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a_0	a_1	a_2	a_3												a_{15}	a_{16}	a_{17}

$$\sum_{i=0}^{17} a_i$$

Summation

The symbol $\sum$ converts a sequence of numbers into a **sum**

$$\sum_{i=\ell}^m a_i$$

pronounced as “sum of a_i , where i goes from ℓ to m ”

- i : index of summation
- ℓ, m : lower & upper limit

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a_0	a_1	a_2	a_3							a_{10}					a_{15}	a_{16}	a_{17}

$$\sum_{i=2}^{10} a_i$$

Summation

$$0 + 1 + 2 + 3 + 4 + \dots + n$$

$$\sum_{i=0}^n i$$

```
sum = 0;
for(i=0; i<=n; i++)
{
    sum = sum + i;
}
```

First n terms of

$$0 + 3 + 8 + 15 + 24 + \dots$$

$$\sum_{i=1}^n i^2 - 1$$

```
sum = 0;
for(i=1; i<=n; i++)
{
    sum = sum + i * i - 1;
}
```

Summation

Write the following in expanded form

$$\sum_{i=3}^6 2i^2 = 2(3^2) + 2(4^2) + 2(5^2) + 2(6^2)$$

$$\sum_{n=4}^1 n^3 = 4^3 + 3^3 + 2^3 + 1^3$$

$$\boxed{\text{ICP 7-7}} \quad \sum_{i=1}^3 (i^2 + 2i + 1) = \sum_{i=1}^3 (i + 1)^2 = 2^2 + 3^2 + 4^2$$

$$\boxed{\text{ICP 7-8}} \quad 2 \sum_{j=6}^3 j^2 = 2(6^2) + 2(5^2) + 2(4^2) + 2(3^2)$$

Summation: Change of variable

Changing the variables sometimes helps evaluating the sum by bringing it into a familiar form

$$\sum_{j=1}^6 (j+3)^3 = 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3$$

Let $k = j + 3$ then

$$\sum_{j=1}^6 (j+3)^3 = \sum_{k=4}^9 k^3$$

Be careful in evaluating the lower and upper limits

Linearity of Summation

A function $f(\cdot)$ is linear if for all real numbers a and b

$$f(ax + by) = a \cdot f(x) + b \cdot f(y)$$

Summation is linear!

$$\sum_{i=1}^n (p \cdot a_i + q \cdot b_i) = p \cdot \sum_{i=1}^n a_i + q \cdot \sum_{i=1}^n b_i$$

In the simpler form ($q = 0$), we get

$$\sum_{i=1}^n p \cdot a_i = p \cdot \sum_{i=1}^n a_i$$

Linearity of Summation

$$\sum_{i=1}^3 7i^2 = 7(1)^2 + 7(2)^2 + 7(3)^2 = 7(1^2 + 2^2 + 3^2) = 7 \sum_{i=1}^3 i^2$$

$$\sum_{i=1}^n p \cdot a_i = p \cdot a_1 + p \cdot a_2 + \dots + p \cdot a_n$$

$$= \underbrace{a_1 + \dots + a_1}_{p \text{ times}} + \underbrace{a_2 + \dots + a_2}_{p \text{ times}} + \dots + \underbrace{a_n + \dots + a_n}_{p \text{ times}}$$

$$= \underbrace{(a_1 + \dots + a_n) + (a_1 + \dots + a_n) + \dots + (a_1 + \dots + a_n)}_{p \text{ times}}$$

$$= \underbrace{\sum_{i=1}^n a_i + \sum_{i=1}^n a_i + \dots + \sum_{i=1}^n a_i}_{p \text{ times}} = p \cdot \sum_{i=1}^n a_i$$

Sum Applications

When **Friedrich Gauss** was nine years old, their teacher wanted to pass time so asked the class to evaluate

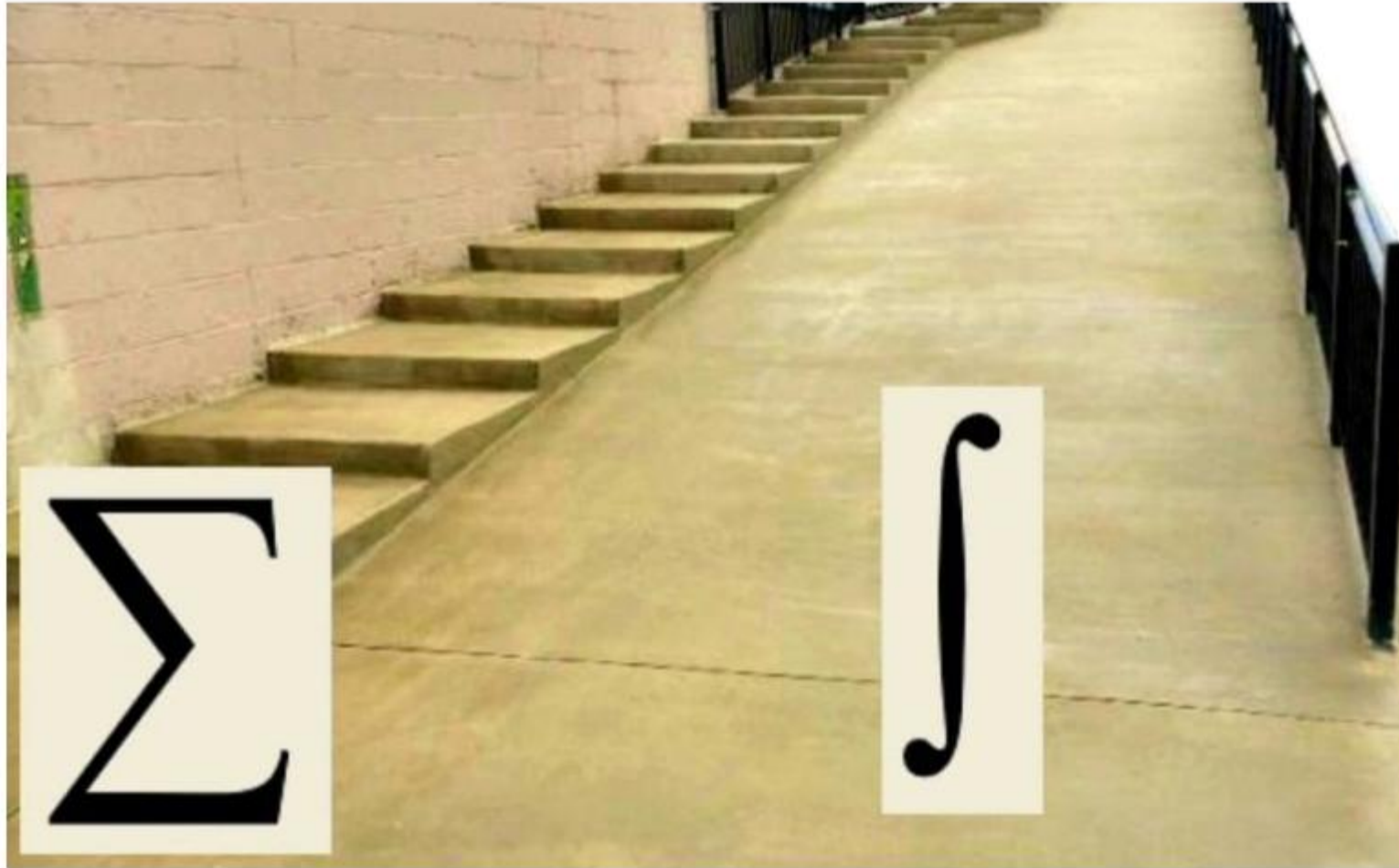
$$1 + 2 + 3 + \dots + 999 + 1000$$

Gauss solved the problem in 30 seconds

It seems a lot of stuff is attributed to Gauss – either he was really smart or he had a great press agent.

Sidenote in 'Concrete Mathematics'

Summation is the discrete analog of Integration



Product

Occasionally, we use the product.

The symbol $\prod$ converts a sequence of numbers into a **product**

$$\prod_{i=0}^n a_i = a_0 \times a_1 \times a_2 \times \cdots \times a_n$$

pronounced as “product of a_i , where i goes from 0 to n ”

```
answer = 1;
for(i=0; i<=n; i++)
{
    answer = answer * a[i];
}
```

Double or Nested Summation

For a sequence of numbers $a_{i,j}$ (a sequence indexed by two variables) or 2D array or matrix, double summation converts it into a **sum**

$$\begin{aligned} & \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} a_{i,j} \\ &= a_{1,1} + a_{1,2} + \cdots + a_{1,n_2} \\ &+ a_{2,1} + a_{2,2} + \cdots + a_{2,n_2} \\ &+ a_{3,1} + a_{3,2} + \cdots + a_{3,n_2} \\ &\vdots \\ &\vdots \\ &+ a_{n_1,1} + a_{n_1,2} + \cdots + a_{n_1,n_2} \end{aligned}$$

	1	2	3		n_2
1	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	$\cdot \cdot \cdot$	a_{1,n_2}
2	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$	$\cdot \cdot \cdot$	a_{2,n_2}
3	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	$\cdot \cdot \cdot$	
	$\ddots$				
n_1	$a_{n_1,1}$	$a_{n_1,2}$	$a_{n_1,3}$	$\cdot \cdot \cdot$	a_{n_1,n_2}

Double or Nested Summation

For a sequence of numbers $a_{i,j}$ (a sequence indexed by two variables) or 2D array or matrix, double summation converts it into a **sum**

$$\begin{aligned} & \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} a_{i,j} \\ &= a_{1,1} + a_{1,2} + \cdots + a_{1,n_2} \\ &+ a_{2,1} + a_{2,2} + \cdots + a_{2,n_2} \\ &+ a_{3,1} + a_{3,2} + \cdots + a_{3,n_2} \\ &\vdots \\ &\vdots \\ &+ a_{n_1,1} + a_{n_1,2} + \cdots + a_{n_1,n_2} \end{aligned}$$

	1	2	3	...	n_2
1	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	...	a_{1,n_2}
2	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$	...	a_{2,n_2}
3	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	...	
...					
n_1	$a_{n_1,1}$	$a_{n_1,2}$	$a_{n_1,3}$	...	a_{n_1,n_2}

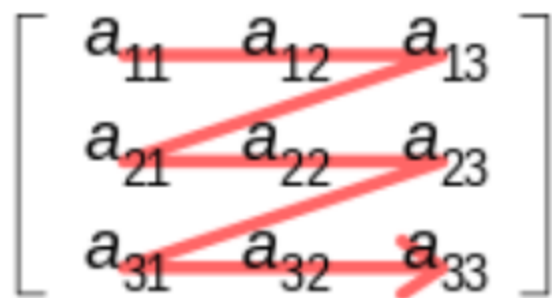
```
sum = 0;
for(i=1; i<=n1; i++)
{
    for(j=1; j<=n2; j++)
    {
        sum = sum + a[i][j];
    }
}
```

Double or Nested Summation

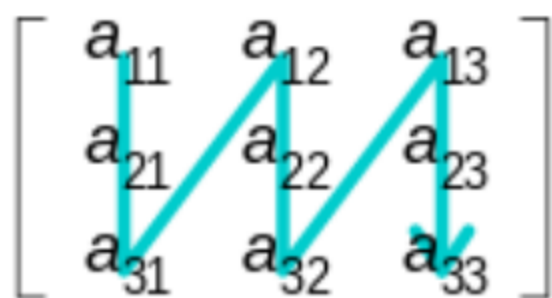
For a sequence of numbers $a_{i,j}$ (a sequence indexed by two variables) or 2D array or matrix, double summation converts it into a **sum**

$$\sum_{i=1}^{n_1} \sum_{j=1}^{n_2} a_{i,j} = \sum_{j=1}^{n_2} \sum_{i=1}^{n_1} a_{i,j}$$

Row-major order



Column-major order



Double or Nested Summation

For a sequence of numbers $a_{i,j}$ (a sequence indexed by two variables) or 2D array or matrix, double summation converts it into a **sum**

$$\sum_{i=1}^{n_1} \sum_{j=1}^i a_{i,j}$$

$$= a_{1,1}$$

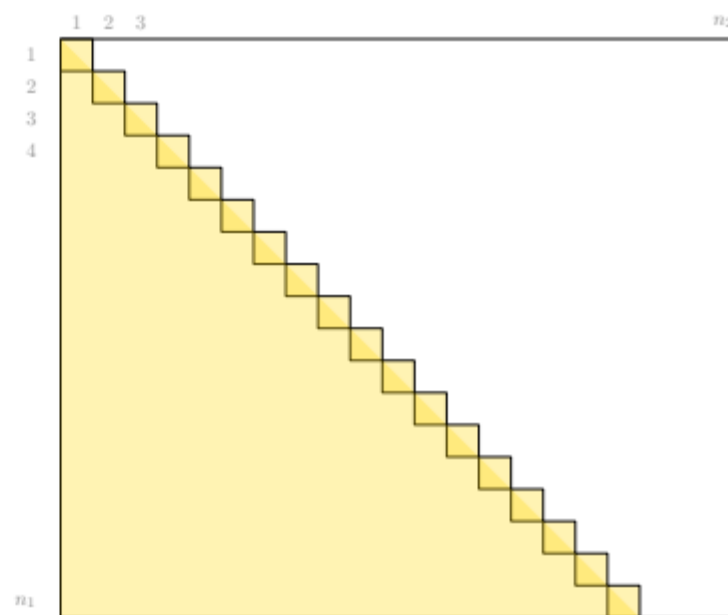
$$+ a_{2,1} + a_{2,2}$$

$$+ a_{3,1} + a_{3,2} + a_{3,3}$$

$$+ a_{4,1} + a_{4,2} + a_{4,3} + a_{4,4}$$

$$\vdots$$

$$+ a_{n_1,1} + a_{n_1,2} + \cdots + a_{n_1,n_1}$$



Assuming $n_1 \leq n_2$

Double or Nested Summation

For a sequence of numbers $a_{i,j}$ (a sequence indexed by two variables) or 2D array or matrix, double summation converts it into a **sum**

$$\sum_{i=1}^{n_1} \sum_{j=1}^i a_{i,j}$$

$$= a_{1,1}$$

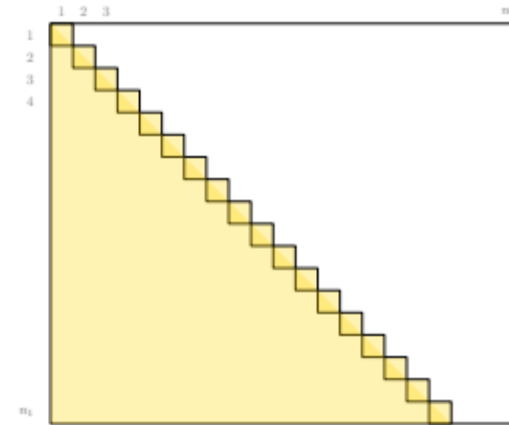
$$+ a_{2,1} + a_{2,2}$$

$$+ a_{3,1} + a_{3,2} + a_{3,3}$$

$$+ a_{4,1} + a_{4,2} + a_{4,3} + a_{4,4}$$

$$\vdots$$

$$+ a_{n_1,1} + a_{n_1,2} + \cdots + a_{n_1,n_1}$$



```
sum = 0;
for(i=1; i<=n1; i++)
{
    for(j=1; j<=i; j++)
    {
        sum = sum + a[i][j];
    }
}
```

Assuming $n_1 \leq n_2$

Evaluating Sums

Given a summation find a *closed form* formula to evaluate it

A formula involving lower and upper limits only, to output value of the sum

For instance what is the value (in terms of n) of $\sum_{i=1}^n i$?

Evaluating Sums

Evaluate $T_n := \sum_{i=1}^n i = 1 + 2 + 3 + \dots + n - 2 + n - 1 + n$

$$\begin{array}{c} \overbrace{\hspace{10em}}^{n+1} \\ \overbrace{\hspace{8em}}^{n+1} \\ \overbrace{\hspace{6em}}^{n+1} \\ 1 + 2 + 3 + \dots + (n-2) + (n-1) + n \end{array}$$

- Each pair sums up to $n + 1$
- Number of pairs is $n/2$ ▷ Think about n odd/even

$$T_n = \frac{n(n+1)}{2}$$

Evaluating Sums

$$\text{Evaluate } T_n := \sum_{i=1}^n i = 1 + 2 + 3 + \dots + n - 2 + n - 1 + n$$

An alternative way to evaluate T_n

$$\begin{array}{r|cccccccc} T_n = & 1 & + & 2 & + & \dots & + & n-1 & + & n \\ T_n = & n & + & n-1 & + & \dots & + & 2 & + & 1 \\ \hline 2T_n = & n+1 & + & n+1 & + & \dots & + & n+1 & + & n+1 \end{array}$$

$$2T_n = n(n+1) \quad \Rightarrow \quad T_n = \frac{n(n+1)}{2}$$

Evaluating Sums

Evaluate $E_n := 2 + 4 + 6 + \dots + 2(n-2) + 2(n-1) + 2n$

This is

$$E_n := \sum_{i=1}^n 2i$$

By linearity of summation

$$E_n = 2 \sum_{i=1}^n i = 2T_n = 2 \frac{n(n+1)}{2} = n(n+1)$$

Square Numbers

We express square numbers in terms of T_n

$$n^2 = T_n + T_{n-1}$$

This can be easily verified as

$$T_n + T_{n-1} = \frac{n(n+1)}{2} + \frac{(n-1)n}{2} = n^2$$

Sum of odd numbers

Evaluate

$$O_n := \sum_{i=1}^n (2i - 1) = 1 + 3 + 5 + \dots + (2n - 3) + (2n - 1)$$

By linearity of summation

$$O_n = 2 \sum_{i=1}^n i - \sum_{i=1}^n 1 = 2T_n - n = 2 \frac{n(n+1)}{2} - n = n(n+1) - n$$

$$O_n = 1 + 3 + 5 + \dots + 2(n-2) - 1 + 2(n-1) - 1 + 2n - 1 = n^2$$

Sum of odd numbers

We got $O_n := 1 + 3 + \dots + 2n - 3 + 2n - 1 = n^2$

Our earlier expression: $n^2 = T_n + T_{n-1} = \sum_{i=1}^n i + \sum_{i=1}^{n-1} i$

$$O_n = \overbrace{1 + 2 \dots + (n-1) + n}^{T_n} + \overbrace{1 + 2 \dots + (n-2) + (n-1)}^{T_{n-1}}$$

Another way to look at this is

T_{n-1}				1	+	2	+	3	+	4	+	5	...
T_n		1	+	2	+	3	+	4	+	5	+	6	...
O_n		1	+	3	+	5	+	7	+	9	+	11	...

Sum of squares and cubes

Evaluate sum of squares and sum of cubes of first n positive integers

$$S_n := \sum_{i=1}^n i^2 := 1^2 + 2^2 + 3^2 + \dots + n^2$$

and

$$C_n := \sum_{i=1}^n i^3 := 1^3 + 2^3 + 3^3 + \dots + n^3$$

Sum of Squares

Evaluate

$$S_n := \sum_{i=1}^n i^2 = 1^2 + 2^2 + 3^2 + \dots + n^2$$

Sum of Squares

$$(a + 1)^3 = a^3 + 3a^2 + 3a + 1$$

1^3	$=$	$(0 + 1)^3$	$=$	0^3	$+$	$3 \cdot 0^2$	$+$	$3 \cdot 0$	$+$	1
2^3	$=$	$(1 + 1)^3$	$=$	1^3	$+$	$3 \cdot 1^2$	$+$	$3 \cdot 1$	$+$	1
3^3	$=$	$(2 + 1)^3$	$=$	2^3	$+$	$3 \cdot 2^2$	$+$	$3 \cdot 2$	$+$	1
4^3	$=$	$(3 + 1)^3$	$=$	3^3	$+$	$3 \cdot 3^2$	$+$	$3 \cdot 3$	$+$	1
$\vdots$		$\vdots$		$\vdots$		$\vdots$		$\vdots$		$\vdots$
$(n + 1)^3$	$=$	$(n + 1)^3$	$=$	n^3	$+$	$3 \cdot n^2$	$+$	$3 \cdot n$	$+$	1
$\sum$		$\sum_{i=0}^n (i + 1)^3$	$=$	$\sum_{i=0}^n i^3$	$+$	$3 \sum_{i=0}^n i^2$	$+$	$3 \sum_{i=0}^n i$	$+$	$n + 1$

Sum of Squares

$$\sum_{i=0}^n (i+1)^3 = \sum_{i=0}^n i^3 + 3 \sum_{i=0}^n i^2 + 3 \sum_{i=0}^n i + n + 1$$

$$\sum_{i=1}^{n+1} i^3 = \sum_{i=1}^n i^3 + 3 \sum_{i=0}^n i^2 + 3 \sum_{i=0}^n i + n + 1$$

$$\cancel{\sum_{i=1}^n i^3} + (n+1)^3 = \cancel{\sum_{i=1}^n i^3} + 3 \sum_{i=0}^n i^2 + 3 \sum_{i=0}^n i + n + 1$$

S_n T_n

$$\Rightarrow 3S_n = 3 \sum_{i=1}^n i^2 = (n+1)^3 - 3 \sum_{i=0}^n i - (n+1)$$

$$\Rightarrow S_n = \sum_{i=1}^n i^2 = (n+1)(n+1/2)n = \frac{n(n+1)(2n+1)}{6}$$

Sum of Cubes

Evaluate $C_n := \sum_{i=1}^n i^3 = 1^3 + 2^3 + 3^3 + \dots + n^3$

$$1^3 + 2^3 + \dots (n-1)^3 + n^3 = (T_n)^2$$

$$= \left(\frac{n(n+1)}{2} \right)^2$$

Important Sums

$$\sum_{i=1}^n i = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n 2i = 2 + 4 + \dots + 2n = n(n+1)$$

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$\sum_{i=1}^n 2i - 1 = 1 + 3 + \dots + 2n - 1 = n^2$$

Evaluating Sums

Evaluate the following sum in terms of appropriate variables

$$\sum_{i=3}^n 2(i-1)^2$$

Let $j = i - 1$, we get

$$\sum_{j=2}^{n-1} 2j^2 = 2 \sum_{j=2}^{n-1} j^2 = 2 \left(\sum_{j=1}^{n-1} j^2 - 1^2 \right) = 2 \left(\frac{(n-1)(n)(2n-1)}{6} - 1^2 \right)$$

Evaluating Sums

Evaluate the following sum in terms of appropriate variables

$$\sum_{n=k}^1 3n^3 = 3 \sum_{n=k}^1 n^3 = 3 \sum_{n=1}^k n^3 = 3 \left(\frac{k(k+1)}{2} \right)^2$$

Evaluating Sums

Evaluate the following sum in terms of appropriate variables

ICP 7-9

$$\sum_{k=1}^n (k^2 + 2k + 1)$$

ICP 7-10

$$\sum_{i=1}^m 3i^3 + 4i^2$$

Telescoping Sum

Let $\{a_i\}$ be a sequence. Evaluate

$$\sum_{i=0}^n a_{i+1} - a_i$$

$$\sum_{i=0}^n a_{i+1} - a_i$$

$$= (a_1 - a_0) + (a_2 - a_1) + (a_3 - a_2) + \cdots + (a_n - a_{n-1}) + (a_{n+1} - a_n)$$

$$= (\cancel{a_1} - a_0) + (a_2 - \cancel{a_1}) + (a_3 - a_2) + \cdots + (a_n - a_{n-1}) + (a_{n+1} - a_n)$$

$$= (\cancel{a_1} - a_0) + (\cancel{a_2} - \cancel{a_1}) + (\cancel{a_3} - \cancel{a_2}) + \cdots + (\cancel{a_n} - \cancel{a_{n-1}}) + (a_{n+1} - \cancel{a_n})$$

$$= a_{n+1} - a_0$$

Telescoping Sum

Evaluate $\sum_{k=1}^n \frac{1}{k(k+1)}$

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

$$\begin{aligned}\sum_{k=1}^n \frac{1}{k(k+1)} &= \sum_{k=1}^n \frac{1}{k} - \frac{1}{k+1} \\&= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\&= \left(\frac{1}{1} - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}}\right) + \cdots + \left(\cancel{\frac{1}{n-1}} - \cancel{\frac{1}{n}}\right) + \left(\cancel{\frac{1}{n}} - \frac{1}{n+1}\right) \\&= \frac{1}{1} - \frac{1}{n+1} = \frac{n}{n+1}\end{aligned}$$

Evaluating Sums

- Given a summation find a *closed form* formula to evaluate it
- A formula to output value of the sum using the values of limits only

$$\sum_{i=1}^n i = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n 2i = 2 + 4 + \dots + 2n = n(n+1)$$

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$\sum_{i=1}^n 2i - 1 = 1 + 3 + \dots + 2n - 1 = n^2$$

Geometric Progressions

A **geometric progression** is a sequence of the form

$$a, ar, ar^2, \dots, ar^i, ar^{i+1} \dots$$

where a and r are real numbers

$$\frac{ar^{i+1}}{ar^i} = r$$

- the ratio of consecutive terms is called the **common ratio**
- a is called the **initial term**
- The next term is obtained by multiplying the previous term with r

Finite Geometric Sums

Evaluating finite geometric sum

$$S_n = \sum_{i=0}^{n-1} ar^i$$

Add one more term

$$S_n + ar^n = \sum_{i=0}^{n-1} ar^i + ar^n = ar^0 + \sum_{i=0}^{n-1} ar^{i+1}$$

$$= ar^0 + r \underbrace{\sum_{i=0}^{n-1} ar^i}_{S_n} = ar^0 + rS_n = a + rS_n$$

$$S_n + ar^n = a + rS_n \implies S_n = \frac{a - ar^n}{1 - r} \quad \text{when } r \neq 1$$

Finite Geometric Sums

$$S_n = \sum_{i=0}^{n-1} ar^i \implies S_n = \frac{a - ar^n}{1 - r} \quad \text{when } r \neq 1$$

Some authors refer to this method as the **perturbation method**.

Generally, it works as follows

- 1 Denote an unknown sum as $S_n = \sum_{i=0}^n a_i$
- 2 $S_{n+1} = S_n + a_{n+1} = a_0 + \sum_{i=1}^{n+1} a_i$ $\triangleright$ separate last and first term
- 3 $S_{n+1} = S_n + a_{n+1} = a_0 + \sum_{j=0}^n a_{j+1}$ $\triangleright$ change of variable
- 4 Try to express $\sum_{j=0}^n a_{j+1}$ in terms of S_n
- 5 If can be expressed as multiple of S_n , solve for S_n to get a closed form

Finite Geometric Sums

ICP 7-11

Use the perturbation method to find a closed form of

$$\sum_{i=0}^n i r^i$$

Infinite Geometric Series

Evaluating infinite geometric sum

$$S = \sum_{i=0}^{\infty} a_i = \lim_{n \rightarrow \infty} \sum_{i=0}^n a_i$$

$$S = \sum_{i=0}^{\infty} ar^i = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} ar^i = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a - ar^n}{1 - r}$$

$$S = \lim_{n \rightarrow \infty} \frac{a - ar^n}{1 - r} = \begin{cases} \frac{a}{1 - r} & \text{if } |r| < 1 \\ \infty & \text{else} \end{cases}$$

Infinite Geometric Series

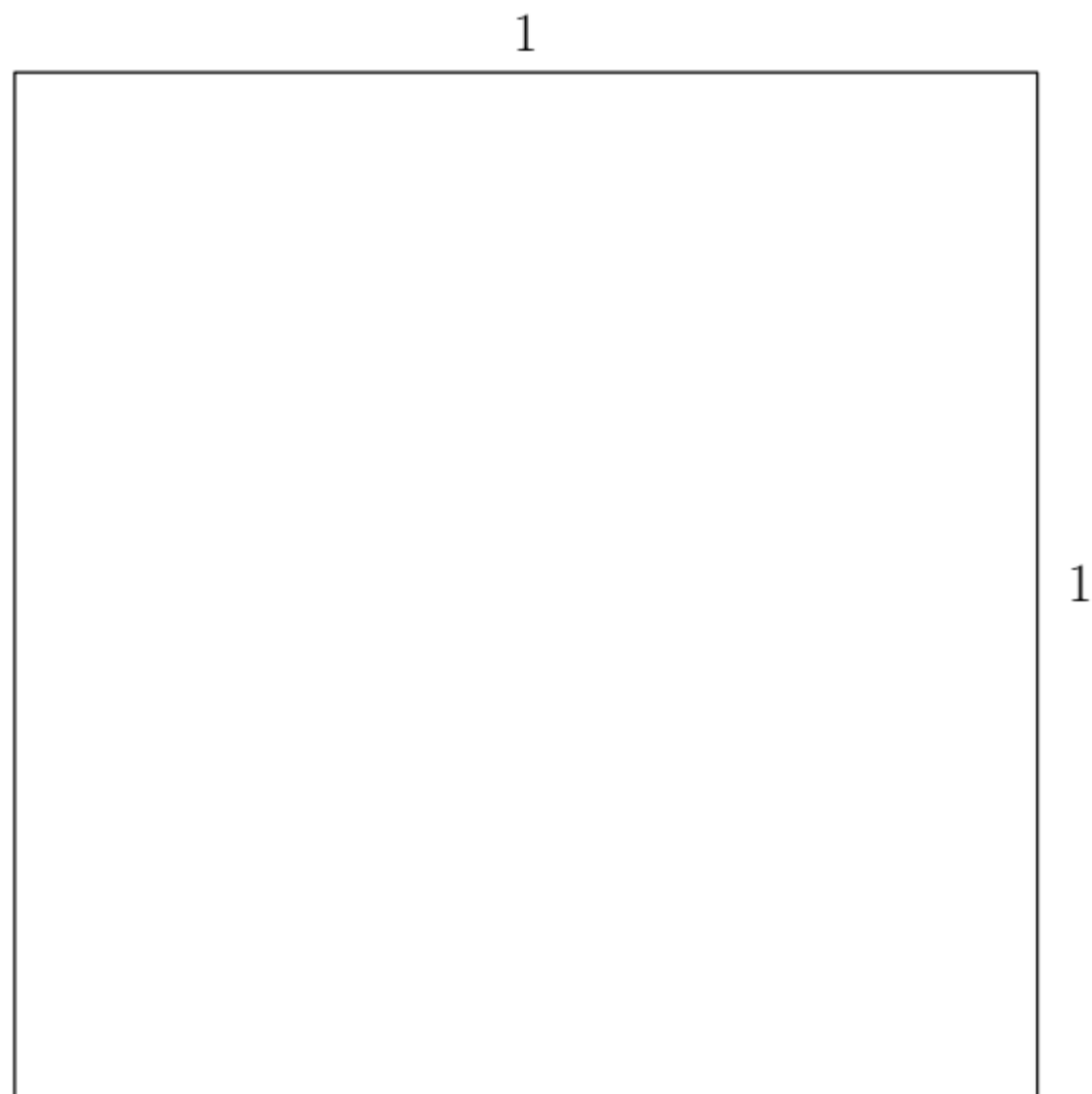
$$S = \sum_{i=0}^{\infty} ar^i = \begin{cases} \frac{a}{1-r} & \text{if } |r| < 1 \\ \infty & \text{else} \end{cases}$$

Evaluate $\sum_{i=0}^{\infty} (1/2)^i$ ($a = 1, r = \frac{1}{2}$)

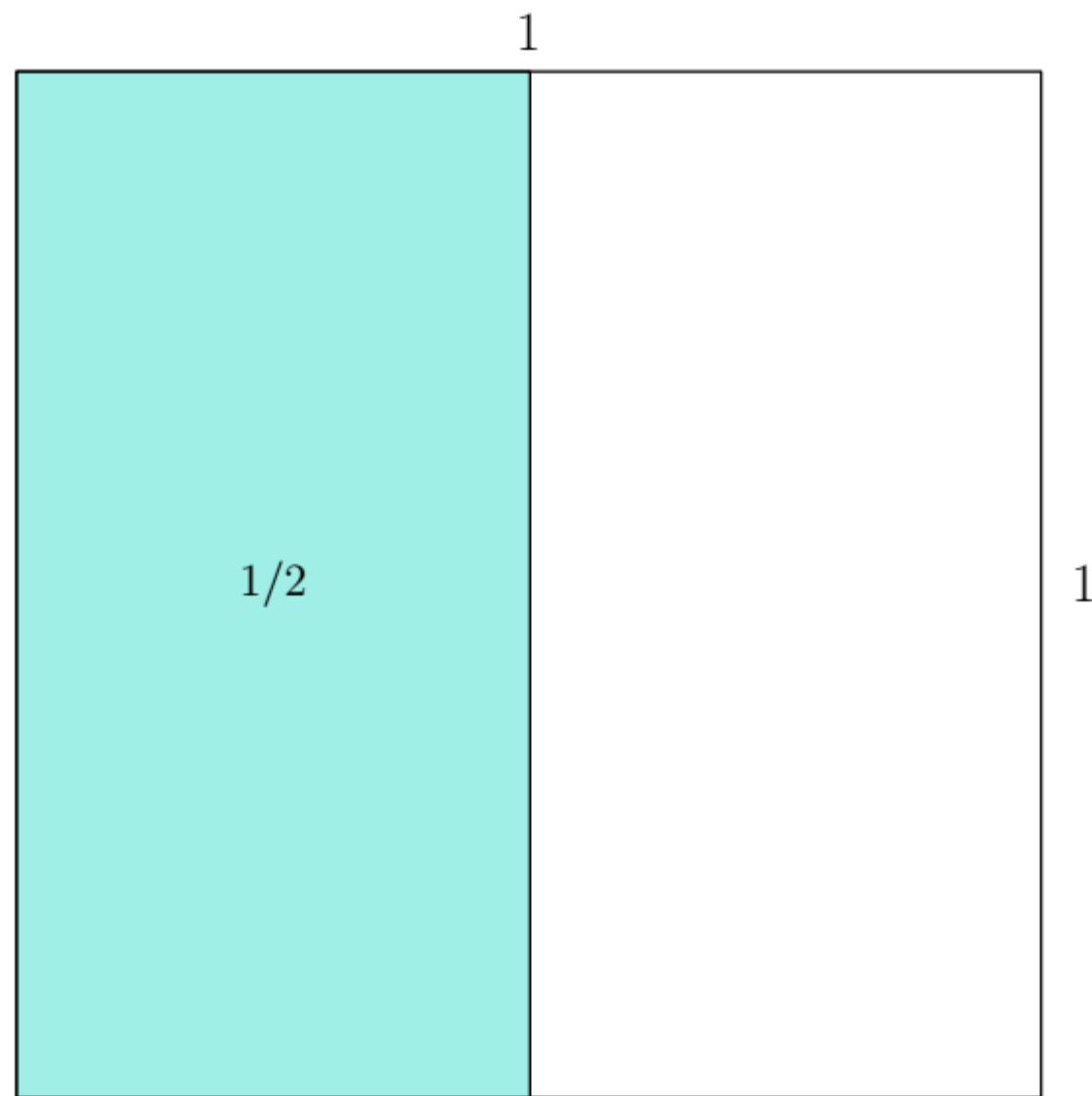
$$\sum_{i=0}^{\infty} \left(\frac{1}{2}\right)^i = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = 2$$

$$\sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots = 1$$

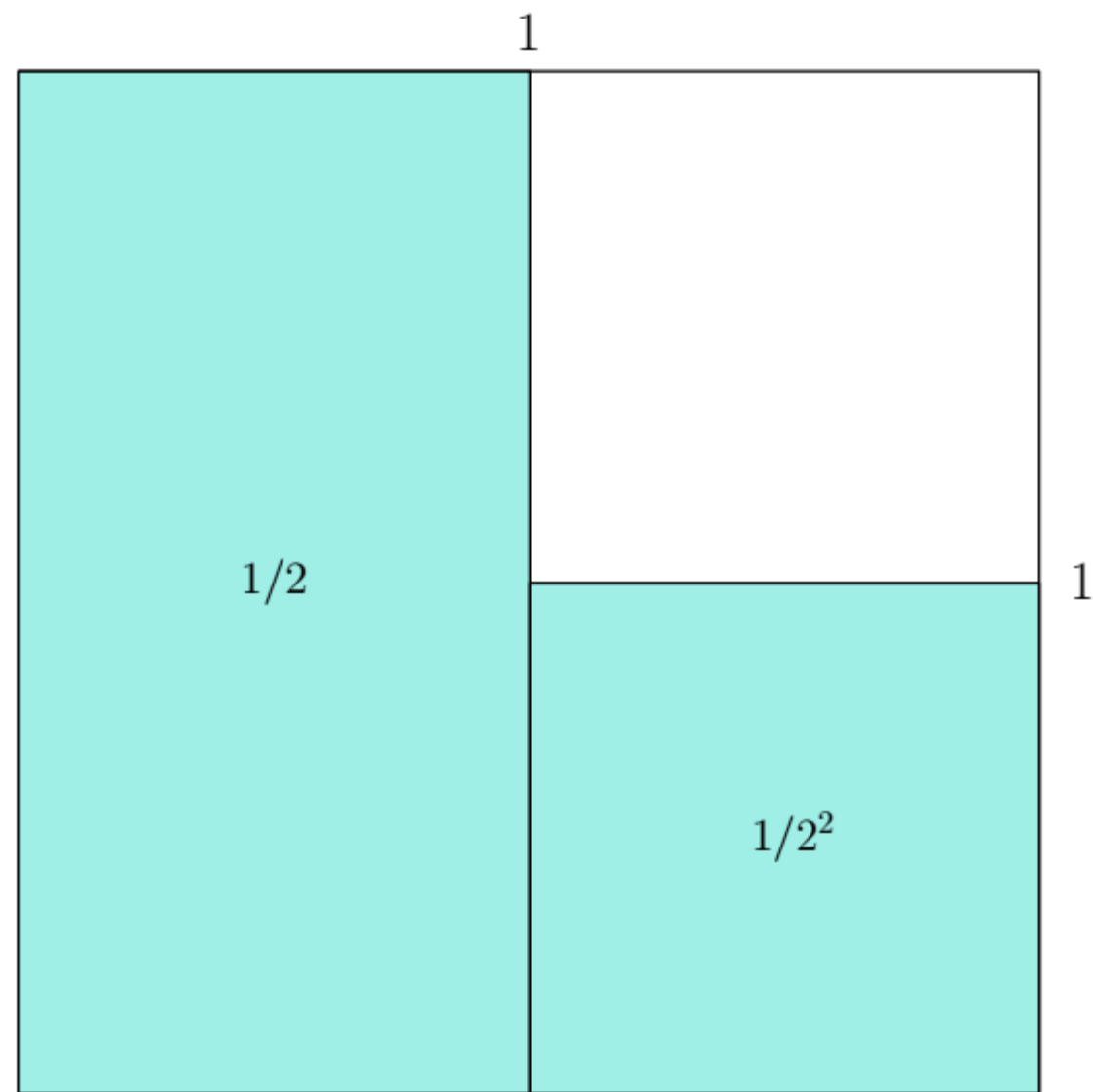
Infinite Geometric Series: Geometric Approach



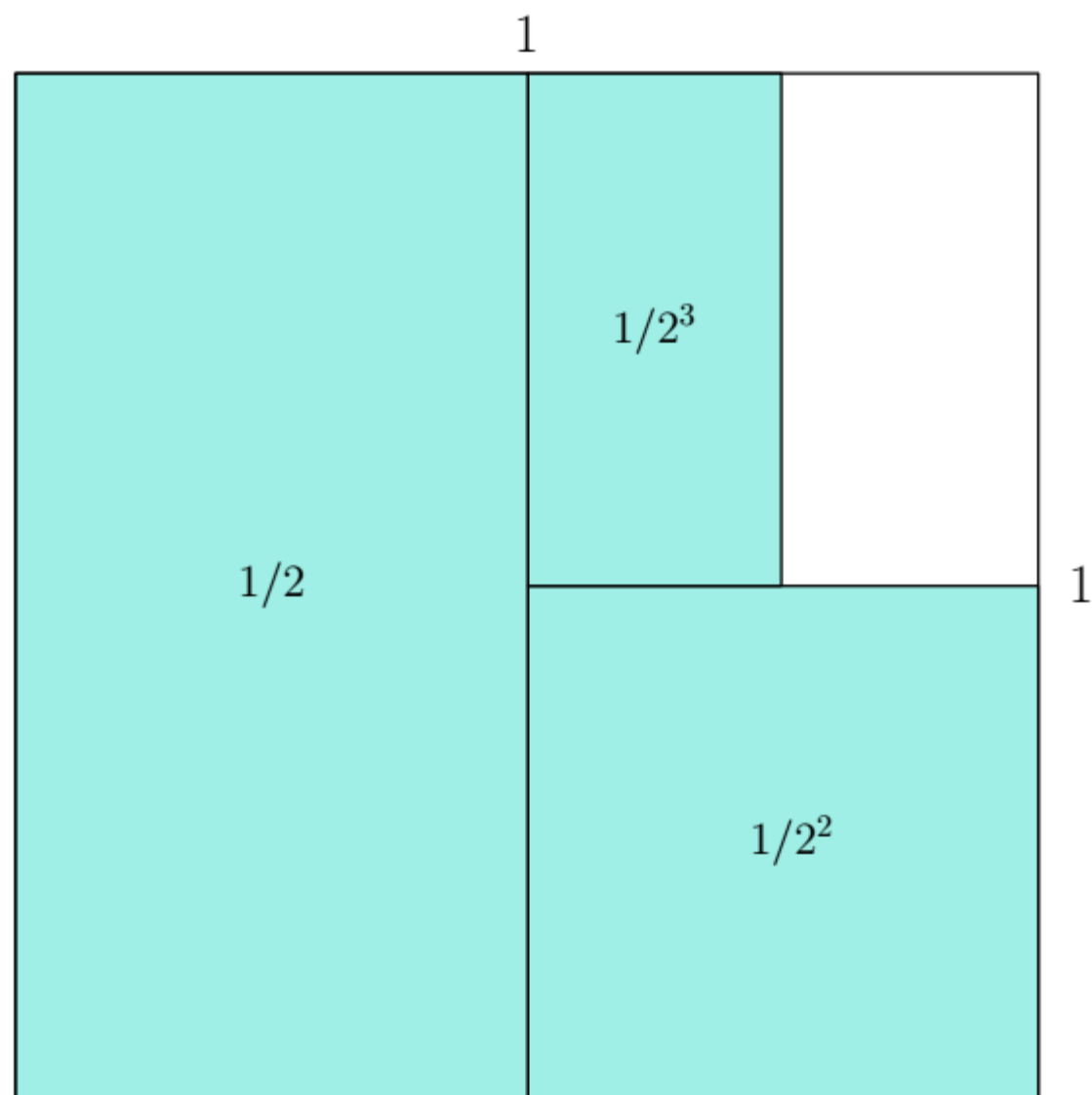
Geometric Sum: Geometric Approach



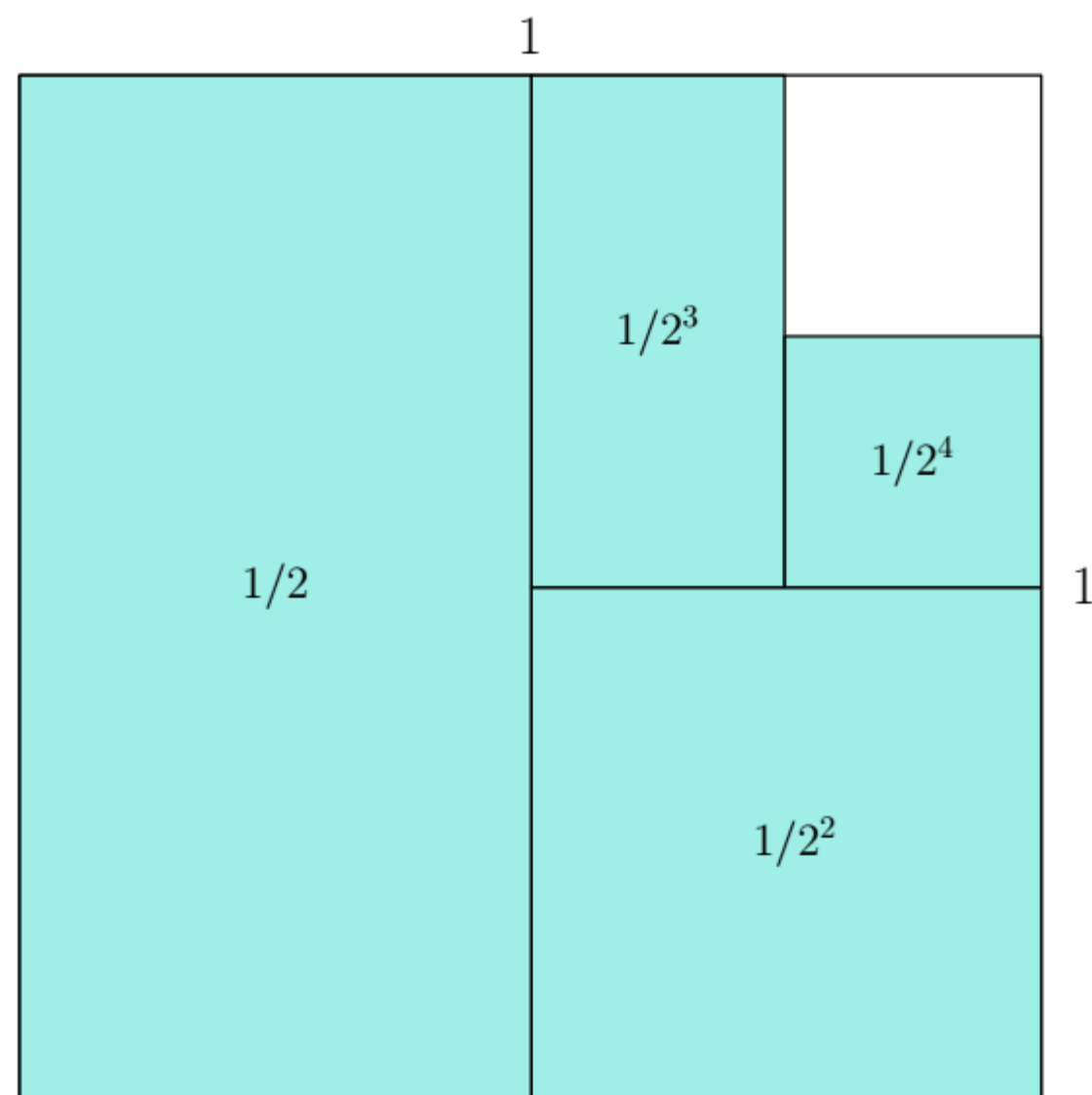
Geometric Sum: Geometric Approach



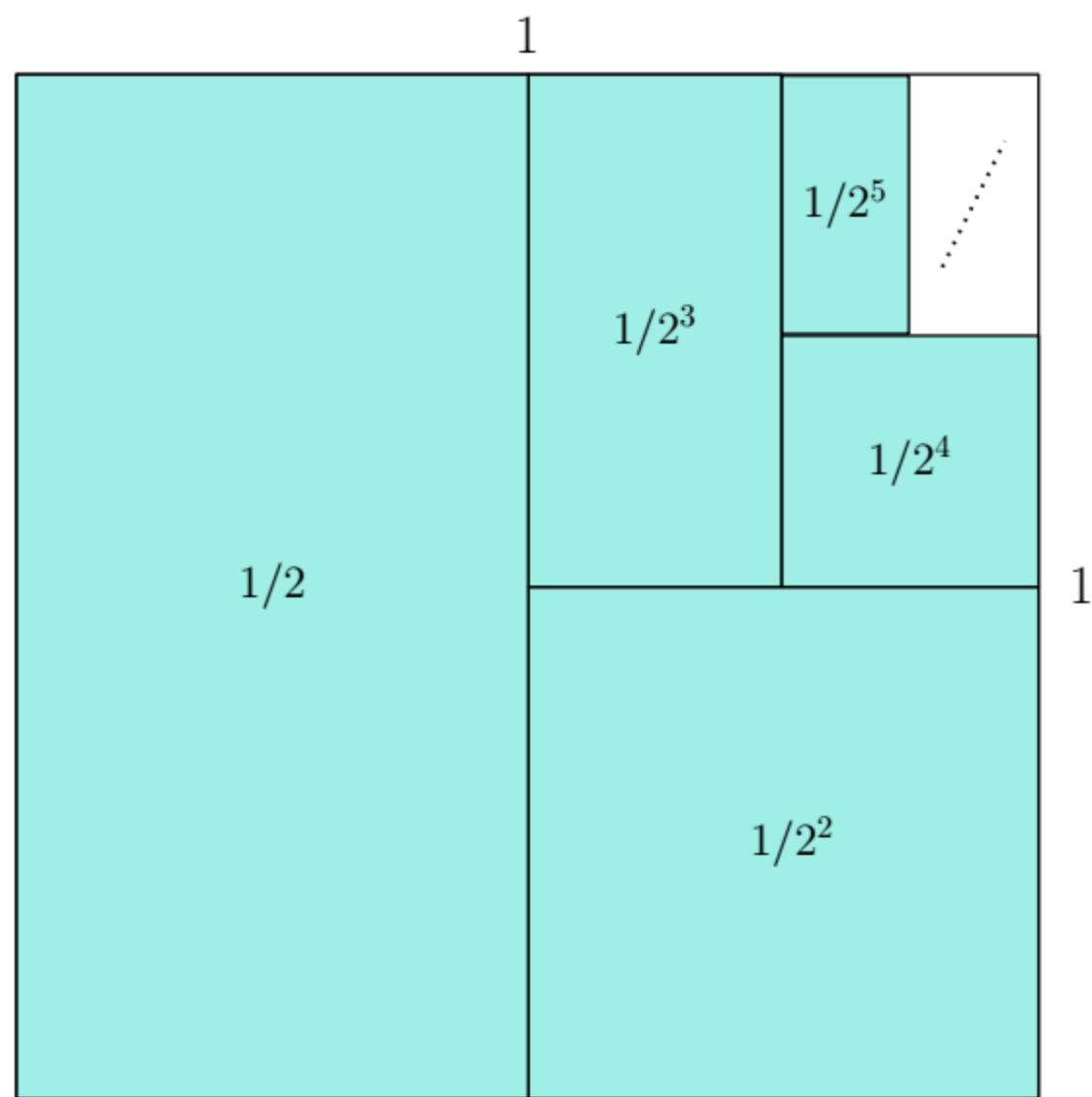
Geometric Sum: Geometric Approach



Geometric Sum: Geometric Approach



Geometric Sum: Geometric Approach



Finite Geometric Sum from Infinite

$$S_n = \sum_{i=0}^{n-1} ar^i = \frac{a - ar^n}{1 - r} \quad \text{when } r \neq 1$$

$$S = \sum_{i=0}^{\infty} ar^i = \begin{cases} \frac{a}{1 - r} & \text{if } |r| < 1 \\ \infty & \text{else} \end{cases}$$

An alternative way to evaluate S_n

$$S = \underbrace{a + ar + ar^2 + \dots + ar^{n-1}}_{S_n} + ar^n + ar^{n+1} + ar^{n+2} + \dots$$

$$S = S_n + ar^n + ar^{n+1} + ar^{n+2} + ar^{n+3} + \dots$$

Finite Geometric Sum from Infinite

An alternative way to evaluate S_n

$$\begin{aligned} S &= \underbrace{a + ar + ar^2 + \dots + ar^{n-1}}_{S_n} + ar^n + ar^{n+1} + ar^{n+2} + \dots \\ &= S_n + ar^n + ar^{n+1} + ar^{n+2} + ar^{n+3} + \dots \\ &= S_n + r^n (a + ar + ar^2 + ar^3 + \dots) \end{aligned}$$

$$S = S_n + r^n S$$

$$\implies S_n = S(1 - r^n)$$

$$\implies S_n = \frac{a(1 - r^n)}{1 - r}$$

Evaluating Sums

Evaluate the following sums in terms of appropriate variables

ICP 7-12

$$\sum_{i=0}^{\infty} 2^i/3^{i+1} = \frac{1}{3} \sum_{i=0}^{\infty} 2^i/3^i = \frac{1}{3} \sum_{i=0}^{\infty} (2/3)^i$$

▷ Infinite geometric sum $a = 1$, $r = 2/3$

ICP 7-13

$$\sum_{i=0}^n 2^i/3^{i+1}$$

ICP 7-14

$$\sum_{i=3}^{\infty} 2^{i+1}/3^i$$

ICP 7-15

$$\sum_{i=2}^n 2^{i+1}/3^i$$